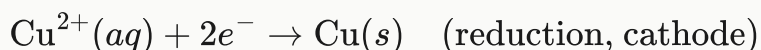
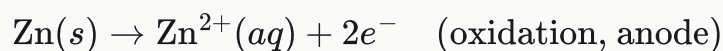


Galvanic Cell

A galvanic cell (also called a voltaic cell) converts chemical energy into electrical energy through a spontaneous oxidation-reduction reaction. The reaction is physically split into two half-cells connected by a wire (for electron flow) and a salt bridge (for ion flow to maintain charge neutrality). At the anode, oxidation occurs and electrons are released; at the cathode, reduction occurs and electrons are consumed. Electrons flow spontaneously through the external wire from anode to cathode, generating a measurable voltage — the cell potential, E_{cell} .

Worked example. Consider the classic zinc-copper cell:



The zinc electrode sits in a ZnSO_4 solution, the copper electrode in a CuSO_4 solution, and a salt bridge (often a KCl-saturated tube) connects the two. Using standard reduction potentials ($E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = +0.34 \text{ V}$, $E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76 \text{ V}$), the standard cell potential is:

$$E_{cell}^{\circ} = E_{cathode}^{\circ} - E_{anode}^{\circ} = 0.34 - (-0.76) = 1.10 \text{ V}$$

This positive value confirms the reaction is spontaneous, consistent with $\Delta G^{\circ} = -nFE_{cell}^{\circ} < 0$.

Problem-solving application. Suppose you're asked to determine whether a silver-nickel cell will run spontaneously, and if so, to identify the anode and predict the voltage. Given $E_{\text{Ag}^{+}/\text{Ag}}^{\circ} = +0.80 \text{ V}$ and $E_{\text{Ni}^{2+}/\text{Ni}}^{\circ} = -0.25 \text{ V}$, the electrode with the higher (more positive) reduction potential — silver — becomes the cathode, and nickel, forced into oxidation, becomes the anode. The cell potential is $0.80 - (-0.25) = 1.05 \text{ V}$, a spontaneous cell.

This same logic underlies real batteries: a AA alkaline cell, a lithium-ion battery, and a car's lead-acid battery are all galvanic cells engineered for stable voltage, rechargeability, or high current output. When troubleshooting a "dead" battery or designing a new cell chemistry, the first question is always the same one this section

answers: which half-reaction is spontaneous, and how do the electrode potentials combine to predict the voltage available to do work?

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